Testing Feedback in QoS Pricing for Decentralized SAGIN Access Association

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***Abstract*—Decentralized access association in a three-tier space–air–ground network must balance throughput with short-term service, because good time averages can conceal a vehicle receiving almost no rate in each slot. Pricing a QoS constraint is a natural remedy, but a multiplier that rises and settles does not by itself establish that feedback caused the gain. We examine this question using per-vehicle expected-shortfall budgets and matched interventions on price updates. Signed aggregation can conceal starvation; one-sided shortfall avoids cancellation without guaranteeing a hard per-slot floor. At $N = 12$, three training seeds show outage reductions of 66.3–86.0% over an unconstrained learner, with throughput and switching costs. Residual shuffling retains the gains, so vehicle-specific update alignment is not established as necessary. Across three budgets, adaptive prices improve budget gaps over a preselected frozen price at .30 and .377, but the stricter target remains unmet; .50 includes a ranking reversal. The results identify a useful service trade-off and delimit the feedback interpretation within the tested SAGIN scenario.**

***Index Terms*—space–air–ground integrated network, multi-agent reinforcement learning, access association, constrained MDP, Lagrangian methods, reproducibility, 6G**

I. Introduction

Sixth-generation networks are expected to integrate low-Earth-orbit (LEO) satellites, high-altitude platform stations and terrestrial base stations into a single space–air–ground integrated network (SAGIN) [1]. No tier dominates for every vehicle at every instant, so vehicles associate dynamically, and when many share the spectrum the association couples: vehicles on the same tier interfere, so a purely greedy rule can congest the strongest layer. A centralized optimizer would need global channel state at vehicular timescales, so we consider *decentralized* policies learned by multi-agent reinforcement learning (MARL).

Driven by throughput alone, such policies can starve poorly served vehicles, and fairness here must be read at *two time scales*: a positive long-run average can hide rotating starvation, while within any one slot some vehicle may still be driven to a near-zero rate. For delay-sensitive traffic [2], instantaneous service matters alongside the long-run average. The natural remedy is to make the floor a constraint and price it with a Lagrangian multiplier updated online by dual ascent. What is rarely checked is whether the resulting price acts as feedback at all: a multiplier that rises and settles is taken as evidence that the constraint is being enforced. This paper asks whether service gains rely on the claimed feedback properties. The training budget is an average specification, not a per-slot guarantee.

A. Adaptation that is not feedback

A price trajectory can look plausible without establishing the mechanism credited with improving service. Signed residuals can cancel across vehicles or time, even when individual slots contain starvation. Conversely, improving throughput and the floor together does not prove an artifact: a finitely trained unconstrained learner need not maximize throughput.

Constraint satisfaction is not self-evidencing. We therefore ask what evidence distinguishes a price being learned from a penalty schedule that happens to work, and answer by intervening on the loop itself. We test two properties separately: whether updates must correspond to the vehicle that incurred a shortfall, and whether adapting prices helps compared with a matched frozen price. A failed explanation need not invalidate a measured service gain.

B. Related work

Learning-based resource management in SAGIN typically relaxes fairness into reward shaping or a time-averaged index. Closest to us, Yang *et al.* [3] apply constraint-aware MADDPG to three-tier uplink scheduling, but their Jain-index constraint becomes a hand-tuned shaped reward and their agents are network-side schedulers over orthogonal sub-channels rather than user-side agents under interference-coupled SINR; Wan *et al.* [4] treat per-UAV QoS in isolation. Pricing has also been used for coordination rather than constraint satisfaction: [5] broadcasts slow-timescale congestion prices, addressing contention rather than a per-vehicle floor.

Our mechanism is an instance of Lagrangian primal–dual safe RL, in which a constraint is relaxed into the reward through a multiplier raised by dual ascent [6], and we adopt its established components rather than propose them: that an expectation constraint does not imply per-step satisfaction, and that one-sided or risk-based costs are the remedy, is standard [7], [8]; that the plain dual is sensitive to initialization and step size is due to [9]; ideal constrained-RL duality results are developed in [10], but do not guarantee convergence of our finite neural multi-agent learner. Our focus is the question:

how does one check that a learned multiplier is functioning as feedback at all? This is the constrained-RL analog of a documented pattern in deep RL, where reported gains were traced to implementation details rather than the mechanism claimed [11]; we apply matched interventions to the price updates in this access problem.

C. Contributions

- **A diagnosis of signed service constraints.** We distinguish shared-residual cancellation from intermittent starvation under feasible per-vehicle expectations. These results motivate shortfall costs without implying that every individual price must remain zero (§III-B).
- **A protocol for testing price feedback.** We intervene on update identity and adaptation using matched residual-shuffle and frozen-price controls, separating service gains from claims about their mechanism (§IV-C).
- **A shortfall-budgeted association method.** We incorporate per-vehicle expected-shortfall budgets into centralized training with decentralized execution, using the same complete residual in the reward and price update. Three budgets expose both benefits and failures of adaptation (§IV-D).
- **The wireless result.** Pricing improves instantaneous minimum service and outage over unconstrained learning across three seeds at $N = 12$, with throughput and switching costs. Historical $N = 6$–20 observations and layer-selection analysis contextualize these gains without asserting large-fleet scalability (§IV-E).

II. System Model

A. Network Architecture

We consider a three-layer SAGIN serving N ground vehicles, each acting as a decision-making agent (Fig. 1). We model *one node per tier*, so association is a choice among tiers rather than among nodes. Separate uplink carriers exclude cross-tier co-channel interference in this modeled scenario: a LEO satellite at 600 km on the Ka-band uplink (30 GHz, NTN reference Set-1 [12]), a high-altitude platform station (HAPS) at 20 km on 1.75 GHz (the range ITU RR footnote 5.388A assigns to HAPS *reception*), and a terrestrial base station on 1.95 GHz (3GPP Band 1 uplink). Each slot, vehicle k selects one tier $a_k^t \in \{\text{LEO}, \text{HAPS}, \text{BS}\}$ for its *uplink*; we study vehicle-side association over access links, not feeder or backhaul selection, in the dense hot-spot regime where all three tiers overlap.

B. Channel and Interference Model

Elevation affects the relative strengths of the tiers; we report at 50°. The two non-terrestrial tiers use free-space path loss with the urban LoS-probability and clutter-loss tables of [13] at the nearest reference elevation, each vehicle drawing its own blockage state; the ground tier uses UMa [14]. Antenna gains and remaining parameters are given in §IV. The satellite link uses the VSAT terminal gain assumption of [12], not a validated moving-terminal link budget. Distances and channels refresh each slot; mobility, Doppler and correlated blockage are not modeled.

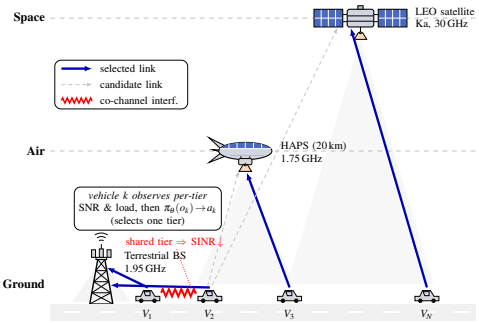


Fig. 1. Vehicle-side association in a three-layer SAGIN. Vehicles sharing a tier interfere on the uplink. Coverage cones and motion arrows are schematic; the simulation uses overlapping coverage and refreshed channel snapshots, not movement trajectories.

Vehicles on the same tier share its spectral resource and interfere on the uplink. With the association indicator and per-tier load

$$x_{k,a}^t = \mathbf{1}\left[a_k^t = a\right], \qquad L_a^t = \sum_{k=1}^{N} x_{k,a}^t, \tag{1}$$

the uplink SINR and achievable rate of vehicle k are

$$\text{SINR}_{k,a}^t = \frac{P\, g_{k,a}^t}{\sum_{j \neq k} x_{j,a}^t P\, g_{j,a}^t + \sigma^2}, \tag{2}$$

$$r_{k,t} = \sum_{a \in \mathscr{A}} x_{k,a}^t \log_2\left(1 + \text{SINR}_{k,a}^t\right), \tag{3}$$

where $g_{k,a}^t$ absorbs path loss, blockage, fading and antenna gains, all vehicles use equal normalized power P, and σ^2 is the noise power. The constrained quantity is the spectral efficiency $r_{k,t}$, which is what the floor and the shortfall budget of §III act on; the delivered rate is $B\, r_{k,t}$ with $B = 20$ MHz identical across tiers, and throughput, floors and outage are reported in Mbps. The load L_a^t enters the denominator of every vehicle on tier a, the coupling that defeats purely greedy selection.

III. Priced Short-Term QoS for Decentralized Access Association

A. Constrained Markov game

Each vehicle k selects a tier every slot from its own observation and receives the spectral efficiency $r_{k,t}$ of (3). Let $\tilde{r}_{k,t}$ be an exponentially weighted moving average of $r_{k,t}$, and let

$$\bar{x}_t \;=\; \alpha \cdot \tfrac{1}{N} \textstyle\sum_j \tilde{r}_{j,t}, \qquad \alpha \in (0,1), \tag{4}$$

be a floor defined *relative* to the fleet. The objective is to maximize mean throughput while keeping every vehicle near $\bar{x}_t$ in *individual slots*. A relative floor keeps one α meaningful as load and geometry change the achievable rate. The EWMA coefficient is $\beta = .9$; EWMAs restart at zero each episode. The reference is policy-dependent, not a fixed Mbps target. Jain's index [15] is not constrained: being scale-invariant, it calls a uniformly starved allocation "fair".

B. What a signed constraint cannot see

The obvious relaxation constrains the rate itself, $g_k = \mathbb{E}[\bar{x}_t - \tilde{r}_{k,t}] \le 0$, and prices it by projected dual ascent. That residual is *signed*: positive when a vehicle is below the floor, negative when it is above. It is not the constraint the objective asks for, and it fails in two separable ways.

Proposition 1 (Signed aggregation cancels the residual)**.** *For $0 < \alpha < 1$ and nonnegative rates,*

$$\tfrac{1}{N}\textstyle\sum_k \big(\bar{x}_t - \tilde{r}_{k,t}\big) \;=\; (\alpha - 1)\cdot \tfrac{1}{N}\textstyle\sum_j \tilde{r}_{j,t} \;\le\; 0. \tag{5}$$

A single multiplier driven by this mean is therefore non-increasing between resets. Reaching zero additionally requires sufficient cumulative negative updates. The cancellation is exact and has nothing to do with learning. The claim is narrow on purpose—it concerns *signed linear* aggregation, and replacing the sum in (5) by $\max_k$, by a mean of hinges, or by a conditional value-at-risk (CVaR) constraint removes the identity.

Per-vehicle multipliers escape Proposition 1, since an individual residual is positive whenever that vehicle is starved, but they fail for a second reason that is a modeling error rather than an algorithmic one. That an expectation constraint does not imply per-step satisfaction is well known in safe RL, and risk measures such as CVaR are the standard response [7], [8].

Proposition 2 (Signed expectations are blind to intermittent starvation)**.** *There are rate processes for which every per-vehicle constraint $\mathbb{E}[\bar{x}_t - r_{k,t}] \le 0$ is satisfied strictly, while in* every *slot some vehicle receives zero rate.*

Proof. Take $N = 2$ with $(r_{1,t}, r_{2,t}) = (0,2)$ for odd t and $(2,0)$ for even t. The fleet mean is 1 every slot, so with $\alpha = 0.25$ each residual is 0.25 half the time and -1.75 the other half, giving $\mathbb{E}[\bar{x}_t - r_{k,t}] = -0.75 < 0$ while each vehicle is starved in half of all slots. Smoothing does not repair this: since $\mathbb{E}[\tilde{r}_k] = \mathbb{E}[r_k]$ in steady state, the EWMA form takes the same value. □

The constraint is thus not failing to be enforced; it permits a service pattern the objective seeks to avoid. This counterexample concerns rates, not a realized price trajectory: a negative-mean residual can still cause repeated increases of a projected per-vehicle multiplier.

C. Budgeting expected shortfall

We instead constrain the *shortfall* the reward already penalizes,

$$\mathbb{E}_\pi\big[\underbrace{(\bar{x}_t - r_{k,t})_+}_{c_{k,t}}\big] \;\le\; \varepsilon\, \mathbb{E}_\pi[\bar{x}_t], \qquad k = 1,\dots,N, \tag{6}$$

with the instantaneous rate inside the hinge, since that is the per-slot event the long-term index hides. One-sided costs are standard [7], [8]; §III-B motivates this remedy without excluding other risk formulations. Non-negative $c_{k,t}$ prevents surplus service from canceling shortfalls. The dimensionless budget ε is readable as *a vehicle may fall short of the floor by at most a fraction ε of it, on average*—and does not guarantee a hard floor in each slot or a particular outage probability. Pricing the rate residual while penalizing the hinge would relax two different constraints, since $\mathbb{E}[(a-r)_+] \ge a - \mathbb{E}[r]$, strictly whenever the residual changes sign.

Because the right-hand side of (6) is itself policy-dependent, the Lagrangian carries it too. With μ weighting association changes, vehicle k's reward and price update are

$$R_{k,t} = \tfrac{1}{N}\textstyle\sum_j r_{j,t} - \lambda_k\big(c_{k,t} - \varepsilon\,\bar{x}_t\big) - \mu\,\mathbf{1}[a_{k,t} \ne a_{k,t-1}], \tag{7}$$

$$\lambda_k \leftarrow \mathrm{clip}\big(\lambda_k + \eta\,\big(c_{k,t} - \varepsilon\,\bar{x}_t\big), 0, \lambda_{\max}\big). \tag{8}$$

The bracketed quantity is the same in both, so the policy is charged exactly the constraint the price tracks; omitting $\varepsilon\bar{x}_t$ from (7) would discard a policy-dependent term. Raising the reference can increase or decrease the residual, depending on whether the vehicle is below it. Rewards use pre-update prices, which persist across episodes within each worker.

D. A budget is a specification; a penalty weight is not

A budget specifies a tolerable shortfall; a penalty coefficient weights that cost during learning. For a *constant* budget c, a fixed price optimizes $R(\pi) - \lambda C(\pi) + \lambda c$, whose last term is independent of π. That algebra does not carry over to (6) unchanged, since our right-hand side is policy-dependent. Even a frozen price can therefore yield different policies at different budgets. Whether adaptation improves budget satisfaction is an empirical question, examined in §IV-D; it is not guaranteed by price motion or a convergence theorem.

E. Fixing the operating budget

A reference policy provides an interpretable calibration quantity:

$$\varepsilon_{\mathrm{ref}} = \frac{\mathbb{E}_{\pi_{\mathrm{greedy}}}\big[\tfrac{1}{N}\textstyle\sum_k (\bar{x}_t - r_{k,t})_+\big]}{\mathbb{E}_{\pi_{\mathrm{greedy}}}[\bar{x}_t]}. \tag{9}$$

The operating value $\varepsilon = .377$ is inherited from the historical calibration, which averaged per-slot ratios; the reported ratio of expectations was .387. These statistics are not interchangeable. We retain .377 as a specified operating point and use ratios of sums for budget evaluation, testing .30 and .50 as sensitivity checks. This does not establish an optimal budget or remove the need to evaluate each vehicle separately.

F. Learning

The priced reward (7) is optimized with a MAPPO-style learner [16] under centralized training and decentralized execution. Observations are three current solo-rate features, three previous tier-load shares, and the previous action; the shared actor takes a one-hot identity input, whose dimension grows with N. Measurements are assumed available without delay or error. (8) runs environment-side at the slot timescale: the price shapes training without entering the observation or deployed policy.

The PPO update uses each vehicle's own advantage, not generally the gradient of a team-average Lagrangian in this interference-coupled game. Policy returns are discounted, whereas price updates and reported budgets are undiscounted.

We assess this implemented learner empirically, without an exact global optimization or convergence claim.

IV. Experiments

A. Setup and protocol

The revised comparisons use $N = 12$ and 200-slot episodes. Uplink carriers are 30, 1.75 and 1.95 GHz for LEO, HAPS and BS at 600 km, 20 km and ground level, at 50° elevation, with 20 MHz bandwidth, normalized power $P = 1$, -100 dBm noise and $81.7/16.5/8.0$ dBi combined antenna gain. Ground distance is uniform on $[100, 800]$ m; fading has exponential power (Nakagami-m with $m = 1$), with log-normal shadowing and blockage. We use $\alpha = .25$, EWMA $\beta = .9$, switching weight $\mu = .05$, price step $\eta = .5$ and cap $\lambda_{\max} = 30$. The actor is a GRU(128), the critic is central, and PPO uses learning rate 3×10^{-4}, clip .2, four epochs, entropy coefficient .001, five-step returns with $\gamma = .99$, and per-agent reward standardization.

We denote intact pricing by I, residual shuffling by Y, and unconstrained learning by U; A/F denote adaptive/frozen prices. I/Y/U use budget .377 and zero initial prices for I/Y. A/F share initial price .486 at budgets .30, .377, and .50. Each condition uses training seeds 0/1/2: 27 policies trained from scratch for 100k environment steps across ten workers. We report the fixed 100k endpoint, not the best checkpoint. Evaluations at 20/40/60/80/100k use 10,000 slots per point: ten blocks of five episodes. Evaluation environments are isolated from training, with separate physical, action-sampling and shuffle random streams. The policy samples actions without learning; prices are not actor inputs at deployment.

For slots $\mathcal{T}$, define

$$q_k = \frac{\sum_{t \in \mathcal{T}} c_{k,t}}{\sum_{t \in \mathcal{T}} \bar{x}_t}, \qquad G_k = q_k - \varepsilon, \qquad Q = \frac{1}{N} \sum_k q_k. \tag{10}$$

All recorded denominators are positive. All vehicle indices meet the pooled empirical budget only if $\max_k G_k \le 0$; $Q \le \varepsilon$ alone is insufficient. Indices are agent positions across episodes, not tracked physical vehicles. We also inspect block-wise gaps. Communication metrics are aggregate throughput, mean instantaneous minimum rate, vehicle outage below .1 Mbps, and switching per vehicle-slot. Switching includes the first action relative to the reset action; the layer analysis excludes these first actions. Three training seeds are the replication unit, not vehicles or evaluation blocks. We report descriptive comparisons, not significance or equivalence. The evaluation pool was reused during development, not an unseen final test. Configurations, source/model hashes and detailed metrics are retained in the reproducibility materials. Historical results below are kept separate.

B. Diagnosing the feedback comparison

The shared cancellation identity does not prove that a per-vehicle multiplier cannot rise. Similarly, a gain in both throughput and minimum rate cannot by itself identify a reset artifact. Historical reset results are not used here to make that causal attribution. The archived yoked implementation used a signed EWMA residual for its price update and omitted the budget term from its reward. It therefore changed more than update identity. The comparisons below instead use the complete residual $d_{k,t} = c_{k,t} - \varepsilon \bar{x}_t$ in both arms.

TABLE I
Matched comparisons and the questions they answer.

Comparison	Question and scope
I/U	Does the priced formulation improve service?
I/Y	Is correct residual-to-vehicle update alignment necessary? Y retains adaptive prices and local rewards.
A/F	Does adapting the price help at matched initialization? F is one preselected price, not a tuned frontier.

C. Testing whether the multiplier is feedback

A curve that rises and settles is not evidence that a price is being learned. We retain intervention-based testing but distinguish two properties rather than require every intervention to destroy the service gain. **Misdirect (Y)** replaces only I's price update with

$$\lambda_{k,t+1} = \mathrm{clip}(\lambda_{k,t} + \eta d_{\rho_t(k),t}, 0, \lambda_{\max}), \tag{11}$$

where ρ_t is a fresh uniform permutation. Its reward still uses vehicle k's own residual. This preserves the residual multiset within Y at each step, not price trajectories between I and Y; some indices may remain unchanged. Y retains adaptation and local service information in the reward. **Cut (F)** holds the price at .486 across slots and resets, while A updates from the same initial price with the same reward. The fixed price is the rounded median of three existing late-training mean prices, selected before A/F runs, not an optimized penalty. I and F are not matched because their initial prices differ. U removes only the QoS price term. Table I summarizes the comparison questions.

At 100k, I improves shortfall, outage and instantaneous minimum over U in all three seeds. Y retains these gains (Fig. 2): I has better service for seed 0, but worse shortfall, outage and instantaneous minimum for seeds 1 and 2. Vehicle-specific update alignment is therefore not established as necessary. This does not prove I/Y equivalence or that all feedback is irrelevant. All six I/Y runs meet every pooled per-vehicle budget, whereas every U run has 12/12 indices over budget. Pooling hides shorter failures: I has 1/1/17 and Y has 1/0/5 over-budget block/index pairs out of 120 for seeds 0/1/2. For I seed 2, the worst pooled gap is $-.0020$ but the worst block gap is $+.0311$; its shortfall also worsens from 60k to 100k.

D. What the shortfall budget delivers, and what it costs

A/F tests whether adaptation helps at matched initialization (Fig. 3). At .30 and .377, A has a smaller worst vehicle gap and lower outage than F in every seed. At .377, A meets all pooled per-vehicle budgets in 3/3 runs and F in 0/3. Outage is 56.6–64.1% lower, with 10.5–22.1% less throughput and 27.7–61.0% more switching than F. A still has 1/1/4 over-budget block/index pairs out of 120.

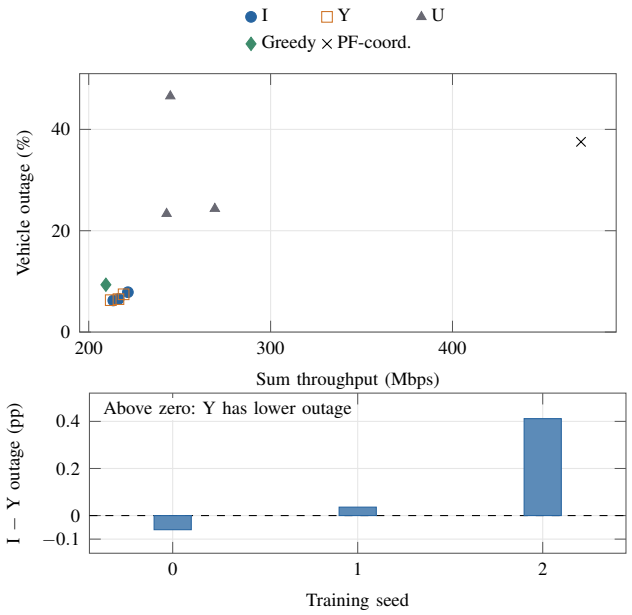


Fig. 2. Service and mechanism comparisons at $N = 12$, 100k steps. Top: each I/Y/U marker is one training seed; each heuristic is one evaluation reference, with PF using global channel information. Bottom: paired I–Y outage differences (percentage points). Shuffling retains the service gains; I has no consistent outage advantage. No fitted frontier is implied.

TABLE II
SERVICE AT $N = 12$, $\varepsilon = .377$: I/Y/U MEANS OVER THREE SEEDS AT 100K; HEURISTICS EACH EVALUATED FOR 10,000 SLOTS. PF USES GLOBAL CHANNEL INFORMATION. SLOT-MIN: MEAN INSTANTANEOUS MINIMUM RATE.

Method	Sum TP (Mbps)	Slot-min (Mbps)	Outage (%)	Switch (%)
I	217.16	0.2097	6.88	53.98
Y	215.88	0.2134	6.75	54.23
U	252.34	0.0116	31.42	30.11
Greedy	209.44	0.1599	9.34	58.51
PF-coordinate	470.66	0.0080	37.54	61.53

The strict .30 budget remains unmet for all twelve indices in every A/F run. A improves the gaps, but its worst gaps remain $+.0347$–$+.0380$ at 100k, with prices reaching the cap of 30. This is failure under the tested settings, not a proof of infeasibility. At .50, each arm passes in 2/3 runs; seed 1 reverses the ordering, with A's worst gap $+.0242$ versus F's $-.0180$. A passes at 20/40/60k but fails at 80/100k; we retain the fixed endpoint. Thus adaptation responds to the budget but does not uniformly meet it or outperform F. These runs neither establish a tuned fixed-penalty frontier nor assign a numerical fraction of benefit to feedback.

E. Access performance

Against U, I reduces mean outage from 31.42% to 6.88% and raises mean instantaneous minimum rate from .0116 to .2097 Mbps (Table II). Paired outage reductions are 66.3–86.0%, at 8.76–20.73% throughput cost and 19.67–29.08 percentage points more switching. This is short-term service redistribution: the minimum time-averaged per-vehicle rate is lower for I (16.86 versus 19.57 Mbps).

Greedy-SNR and PF-coordinate [17] are evaluation references, each using 10,000 slots without training-seed replication. PF-coordinate starts from greedy and makes at most two coordinate sweeps on $\sum_k r_k(\mathbf{a})/\max(m_k, 10^{-6})$, using full current channel information and a separate past-service EWMA m_k. Candidate scoring holds channels and history fixed, updating history once after execution. It is neither globally optimal PF nor a same-information baseline. I has better service than these references, but PF has much higher throughput; there is no universal learned-policy dominance.

Inter-layer selection. Saved endpoint traces show that U's mean tier shares conceal seed-specific concentration: seeds 0/1/2 choose HAPS/BS/LEO in 81.06/82.38/81.56% of vehicle-slots. At .377, A uses LEO in 42.09–44.04% and HAPS in 42.75–44.90% of slots in every seed; F selects HAPS in 59.23–78.73%. I and Y have similar mean shares. Excluding reset actions, A/F switching at .377 is still 54.06/37.90%; LEO–HAPS changes account for 32.08/16.99% of all within-episode transitions. These are associations, not identified causal effects of individual tiers. Switches are selection changes, not physical handovers: no signaling delay or interruption is modeled.

Historical load observations. Archived submission-era CSVs show outage 5.81–8.61% for the priced learner versus 15.84–50.66% for U over $N = 6, 8, 12, 16, 20$ (Fig. 4), with 2.65–14.86% less aggregate throughput. Their mean instantaneous minimum ratios are $17.1\times$ at $N = 12$ and $63.1\times$ at $N = 20$; the latter compares .0480 with .000761 Mbps. These are arithmetic summaries of old runs, not additional runs under the revised protocol. Each load used its own budget, so the trend does not isolate population size. The historical source/checkpoint chain is incomplete, and old models were not re-evaluated. These observations establish neither large-fleet scalability nor the feedback mechanism.

Scope and diagnostic response. The revised evidence is limited to $N = 12$, three seeds and three A/F budgets. When a mechanism test fails, check residuals, reward timing and reset rules before interpreting it; when a budget fails, report the gap and inspect price saturation and checkpoints. This is a diagnostic procedure, not an online recovery controller. The model uses one node per tier, orthogonal tiers and snapshot channels, excluding intra-tier selection, dynamic populations, mobility/Doppler and physical handover costs. Centralized training and assumed load information have unmeasured signaling costs. These are limits of the present conclusions, not verified deployment capabilities.

V. CONCLUSION

A learned multiplier that rises, settles and reports feasibility is not by itself evidence that a constraint is being enforced. In three-tier SAGIN uplink association, signed aggregation and time averaging can conceal starvation. Per-vehicle expected-shortfall budgets retain an interpretable service objective, and

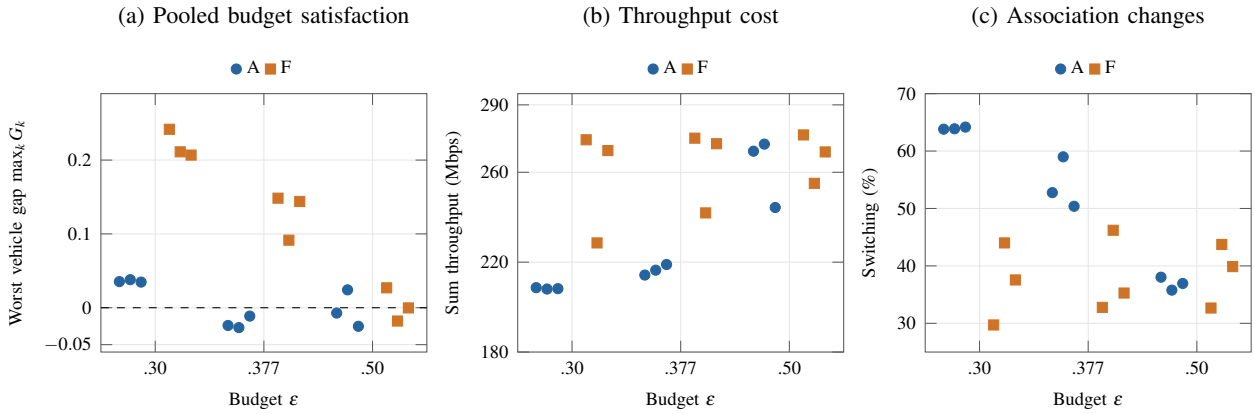


Fig. 3. Matched adaptive/frozen prices (A/F), with the same initial price 0.486, at the fixed 100k endpoint. Each arm shows seeds 0/1/2 from left to right within a budget group; horizontal offsets are only for visibility. All vehicles meet the pooled empirical budget when the worst gap is nonpositive. A improves gaps at .30 and .377, but all strict-budget runs fail. At .50, each arm passes in two of three runs; seed 1 passes only for F. Throughput and switching show the associated costs.

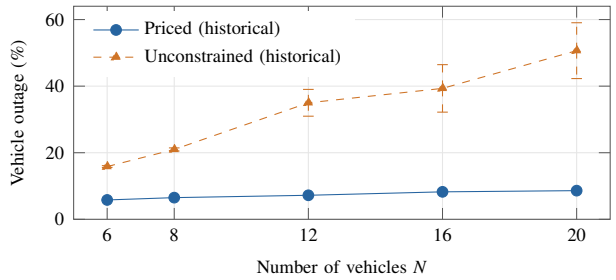


Fig. 4. Historical outage below .1 Mbps versus load. Means and ± 1 standard error over common seed labels: eight at $N = 12$, five elsewhere. Budgets are .336/.361/.377/.383/.379 for increasing N. These old-protocol observations are separate from the revised $N = 12$ comparisons and do not isolate a scaling effect.

the priced learner improves instantaneous minimum service and outage over unconstrained learning, at throughput and switching costs. Matched interventions distinguish that service benefit from its explanation: shuffling retains the gain, while adaptive prices improve budget gaps under some budgets but fail the strict target and exhibit a ranking reversal at the loose one. The contribution is shortfall-budgeted association and an explicit test of its feedback interpretation, not a hard per-slot QoS guarantee or a certification that every feedback component is necessary. The one-node-per-tier snapshot model limits deployment and scaling claims.

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